Partial Identifiability in Discrete Data With Measurement Error

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Abstract

When data contains measurement errors, it is necessary to make assumptions relating the observed, erroneous data to the unobserved true phenomena of interest. These assumptions should be justifiable on substantive grounds, but are often motivated by mathematical convenience, for the sake of exactly identifying the target of inference. We adopt the view that it is preferable to present bounds under justifiable assumptions than to pursue exact identification under dubious ones. To that end, we demonstrate how a broad class of modeling assumptions involving discrete variables, including common measurement error and conditional independence assumptions, can be expressed as linear constraints on the parameters of the model. We then use linear programming techniques to produce sharp bounds for factual and counterfactual distributions under measurement error in such models. We additionally propose a procedure for obtaining outer bounds on non-linear models. Our method yields sharp bounds in a number of important settings – such as the instrumental variable scenario with measurement error – for which no bounds were previously known.

1 Introduction

When data contains measurement errors, it is necessary to account for the errors during analysis in order to avoid biased results. Accounting for measurement error requires making assumptions about how errors occur so as to link the available data to the underlying true values. As such, it is important to the validity of the analysis and the resulting conclusions that these assumptions be substantively justifiable.

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In practice, however, analysts often make implausibly strong assumptions for the sake of exactly identifying their target of inference. We advocate instead for the use of weaker, credible assumptions to identify *bounds* on the target of inference, referred to as *partial identification*. Previous work on partial identifiability under measurement error has largely focused on deriving analytic bounds in an ad-hoc manner for specific combinations of settings and assumptions. As a result, there remain many important and common settings for which no known bounds exist.

In this work, we show how sharp partial identification bounds can be computed numerically and without extensive derivations for a class of models, including multiple settings for which bounds are not currently known. Our approach, which is similar in spirit to that of [1], is to encode the parameter of interest and constraints imposed by the model as a linear program which can be maximized and minimized to produce sharp bounds. We show that this approach can be used to compute bounds for factual and counterfactual parameters of a discrete mismeasured variable, including its marginal distribution and its moments, and the average treatment effect (ATE) on the variable of an intervention.

The primary contribution of this paper is a collection modeling assumptions that can be encoded as linear constraints on a target parameter and, thus, are amenable to the linear programming approach. These include several common **graphical** constraints encoded by hidden variable Bayesian networks, **measurement error** constraints, and **causal** constraints relating potential outcomes under different interventions. In particular, drawing on results from [7] and [5], we show how an extended class of instrumental variable-type models produce linear constraints on a target parameter (Section 3). This class expands the instrumental-style class of graphs long known to have linearly-constrained conditional distributions [7] to include graphs with non-randomized instruments through a result about equivalence in hidden variable Bayesian network models [5]. To facilitate obtaining bounds for graphs outside this class, we present a simple procedure for obtaining

a linear super-model of such graphs. Sharp bounds in the linear super-model will be valid outer-bounds in the original model.

Our approach allows a practitioner to mix and match any combination of the modeling assumptions described in this paper with little effort. This flexibility means that it is trivial to compute bounds under new models and to test the sensitivity of the resulting bounds to different combinations of assumptions. Throughout, we demonstrate the utility of this approach on synthetic examples representing important use cases. Our goal in these examples is to demonstrate how to apply the approach to commonly occurring models so that practitioners may adapt them to their specific data scenario. These cases include:

1. Bounding the mean of a variable using a single noisy proxy
2. Bounding the mean of a variable using two noisy proxies with an exclusion restriction
3. Bounding the ATE in a randomized trial with measurement error on the outcome
4. Bounding the ATE in an instrumental variable model with measurement error on the outcome

These cases represent common scenarios occurring in economics, public health, political science, and other scientific settings. Among these, cases 2, 3, and 4 all represent settings for which no sharp symbolic bounds are currently known and previous work suggests that deriving such symbolic bounds automatically for these settings is infeasible [2]. We begin by introducing our approach in the context of the simplest possible measurement error model, with extensions to other models presented in the following sections.

2 The basic measurement error problem and the LP formulation

Suppose that we are interested in the distribution of a discrete random variable $X \in \mathcal{X}$, but instead of observing X directly, we can only observe a discrete noisy proxy, denoted $Y \in \mathcal{Y}$. Clearly, without any assumptions about the noise distribution, $P(Y|X)$, we cannot say anything about the distribution of X. On the opposite extreme, if $P(Y|X)$ is known and invertible, then $P(X)$ is fully identifiable from observations of Y [17].

Our interest is in between these two extremes, where we assume certain properties of $P(Y|X)$, but not the whole distribution. Our goal is to bound functions of $P(X)$, referred to as *parameters of interest*, under these assumptions. Our approach is to translate the modeling assumptions into a set of constraints on $P(X, Y)$ and bound the parameter of interest by finding its maximum and minimum subject to these constraints. For notational simplicity, let $\phi_{xy} := P(X = x, Y = y)$ be the joint distribution of X and Y. By construction, ϕ must satisfy the the **probability constraints** $\sum_{xy} \phi_{xy} = 1$, and for all x, y, $\phi_{xy} \geq 0$. In addition, ϕ must match the marginal for the observed proxy Y, so that for all y, $\sum_x \phi_{xy} = P(Y = y)$. These are called the **observed data constraints**.

As mentioned above, we must make assumptions about the relationship of the unobserved truth to the observed proxy. We seek to avoid strong, parametric assumptions, in favor of weaker assumptions that may be justified by expert knowledge or domain research. Below, we provide examples of several such commonly made assumptions. We call constraints on ϕ relating the proxies to the unobserved variables the **measurement error constraints** [1].

(A0) Bounded error proportion: $\sum_{x \neq y} \phi_{xy} \leq \epsilon$

(A1) Unidirectional errors: $\sum_{y<x} \phi_{xy} = 0$

(A2) Symmetric error probabilities:
$\phi_{xy} = \phi_{xy} \;\; \forall \; |x - y| = |x - y'|$

(A3) Error probabilities decrease by distance:
$\phi_{xy} \geq \phi_{xy'} \;\; \forall \; |x - y| > |x - y'|$

Note that the target distribution $P(X = x) = \sum_y \phi_{xy}$ and all constraints described thus far can be written as linear functions of ϕ. If our parameter of interest is also a linear function of ϕ, then the assumed constraints, in combination with the linear objective corresponding to our target, form a linear program (LP) and bounds can be computed by both maximizing and minimizing the parameter of interest using one of a number of LP solvers. Parameters that are linear in ϕ include $P(X)$ and all of its moments. By the intermediate value theorem, the objective can take any value between its constrained maximum and minimum values. It follows that for sets of assumptions that can be linearly expressed, the bounds obtained through this linear program are sharp, i.e. they exhaust the information in the observed data under the stated assumptions.

Application: Bounding $E[X]$ using a single proxy

To demonstrate this approach, suppose we are interested in bounding the expected value of a variable

[1] Each of these constraints can easily be soften by adding a slack parameter which can, in turn, be varied in a sensitivity analysis.

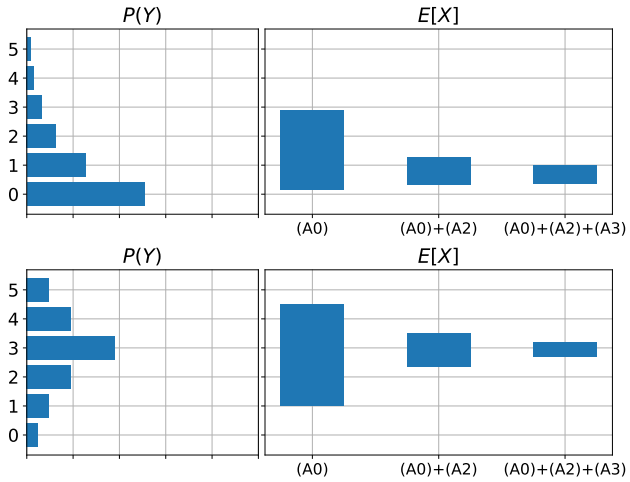

Figure 1: Bounds on $E[X]$ under different measurement error assumptions and observed proxy distributions, $P(Y)$.

$X \in \{0, .., 5\}$ from observations of $Y \in \{0, ..., 5\}$. We will consider bounds under (A2), (A3), and a weaker version of (A0) wherein there is a known upper bound, $\epsilon = 0.01$, on the proportion of observations that are mistaken by more than 2 points (errors of 2 points or less are unconstrained by this assumption). The resulting LP is shown below.

$$\text{objective:} \quad \sum_{x,y} x\phi_{xy} \tag{1}$$

$$\begin{aligned}
\text{constraints:} \quad & \sum_{x} \phi_{xy} = P(Y = y) \\
& \phi_{xy} \geq 0 \\
& \sum_{|x-y|>2} \phi_{xy} \leq \epsilon \\
& \phi_{xy} \geq \phi_{xy'} \qquad \forall |x-y| < |x-y'| \\
& \phi_{xy} = \phi_{xy'} \qquad \forall |x-y| = |x-y'|
\end{aligned}$$

For two example proxy distributions, $P(Y)$, Figure 1 shows the resulting bounds on $E[X]$ under different combinations of (A0), (A2), and (A3). Bounds under each combination of assumptions were computed by simply solving a slightly different version of the LP in Equation 1, highlighting the ease with which we can perform sensitivity analysis without rederiving bounds under each new model. □

Were multiple proxies $\mathbf{Y} \equiv \{Y_1, ..., Y_K\}$ observed with no assumptions made about the relationship between them, each proxy would be subjected to its own observed data constraints and potentially its own measurement error constraints, depending on what knowledge is available about the noise process. The objective, and each of the other constraints, would then simply be expressed on the margins of the full distribution $P(X, \mathbf{Y})$, which maintains linearity.

2.1 Extending the LP approach to other parameters and models

In the remainder of this paper, we show how the LP approach described in this section can be extended to bound other parameters of interest and incorporate other modeling assumptions. In the Section 3, we show how to incorporate conditional independence assumptions, encoded in a graphical model, relating X to other observed and latent variables beyond Y. We define a class of latent variable Bayesian networks which produce linear constraints and show how to relax the constraints imposed by models not in this class to produce valid outer bounds. In Section 4, we consider bounds on parameters of the distribution of X under an intervention A. In this setting, our interest is in the potential outcome variable $X(a)$, defined as the value X would have taken had we intervened to set $A = a$. Our goal, then, is to bound parameters of the distribution $P(X(a))$, including the average treatment effect of A on X. We show how to bound such parameters and incorporate additional assumptions relating potential outcomes under different interventions.

3 Graphical constraints

In the previous section we relied on domain knowledge about the error process through which noisy proxy variables for X were generated. In this section, we describe how assumptions encoded in a graphical model can be used to further constrain our target parameter. In particular, we describe a class of graphs that result in linear constraints on the target parameter and, thus, is amenable to the linear programming approach introduced in the previous section.

Suppose that we assume a latent variable Bayesian network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ represent the vertices and edges of the network, respectively. For a variable $V \in \mathcal{V}$, let $Pa_{\mathcal{G}}(V)$ be the parents of V in $\mathcal{G}$ and $Ch_{\mathcal{G}}(V)$ be the children of V in $\mathcal{G}$. Additionally, we refer to the set of observed and unobserved variables with known cardinality as endogenous variables, denoted by $\mathcal{O}$, and the set of unobserved variables with unknown cardinality (typically latent confounders) as exogenous variables, denoted by $\mathcal{U}$. For example, consider the Bayesian network in Figure 2 (a) in which A, X, and Y are endogenous and U is exogenous. In this model A is commonly referred to as an *instrumental variable* (IV), and the model is referred to as the IV model. The independencies encoded in this graph, namely $A \perp U$ and $A \perp Y \mid A, U$, place constraints on the joint distribution $P(A, U, X, Y)$ which we can used

to bound the target parameter. We refer to the target parameter given by a graphical model as **graphical constraints**.

In general, the independence constraints imposed by a Bayesian network on the joint distribution of variables in the graph are quadratic. For example, the simple Markov chain model shown in Figure 2 (b) yields quadratic constraints on $P(A, X, Y)$. This makes optimizing over the constraint set difficult as, in general, quadratic programming is NP-hard [15]. Further, the complete set of latent variable graphical models that impose linear constraints is unknown; however, in the remainder of this section, we describe a class of graphs, including commonly used graphs such as the IV model, that do yield linear constraints.

To construct this class, we start by considering the IV model shown in Figure 2 (a). In order to arrive at linear constraints, we will not optimize directly over the joint distribution $\phi = P(A, U, X, Y)$ as we did in Section 2. Instead, we will optimize over an equivalent potential outcome distribution. Recall that a potential outcome variable $X(a)$ represents the value X would have taken had we intervened to set $A = a$. Then, let $\tilde{X} = (X(a))_{a\in\mathcal{A}}$ and $\tilde{Y} = (Y(x))_{x\in\mathcal{X}}$ be the vectors of potential outcome variables for X and Y given their endogenous parents. Finally, let ψ be the joint distribution over $\tilde{X}$, and $\tilde{Y}$ such that $\psi_{\tilde{x},\tilde{y}} = P(\tilde{X} = \tilde{x}, \tilde{Y} = \tilde{y})$.

Under the consistency assumption, ψ is connected to the distribution over endogenous variables by the linear map $P(x, y \mid a) = P(X(a) = x, Y(x) = y)$, where the last term is obtained by marginalizing all other variables in $\tilde{X}$ and $\tilde{Y}$ out of the distribution ψ. For an explicit example of this marginalization, see Section 4 in the Supplement. As observed in [2] and [1], all independencies in IV graph are now given by $A \perp \tilde{X}, \tilde{Y}$. As there are no other constraints on ψ, and marginalization is a linear operation, all constraints on the conditional distribution $P(x, y \mid a)$ are linear. This linearity result can be generalized to more complex graphs involving multiple instruments resulting in the following proposition adapted from [7]:

Proposition 1 (Fine's Theorem)**.** *Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a latent variable Bayesian network. Suppose there exists an exogenous latent variable $\Lambda \in \mathcal{V}$ such that (1) all descendants of Λ are children of Λ and (2) all non-descendants of Λ are observed, have exactly one child, and that child is in $Ch_{\mathcal{G}}(\Lambda)$. Let the children of Λ be denoted by $\mathbf{X}$, and the non-descendants of Λ by $\mathbf{A}$. Then all constraints imposed by $\mathcal{G}$ on $P(\mathbf{X} \mid \mathbf{A})$ are linear.*

In such a graph, we refer to all non-descendants of Λ as **instruments** and denote this set by $\mathcal{A}$. We can write distributions in this model as $P(\mathbf{X} \mid \mathbf{A})P(\mathbf{A})$. In our settings, $P(\mathbf{A})$ is assumed to be known since $\mathbf{A}$ is observed, so constraints on the joint distribution of instruments imposed by the model will not play a role in bounding our targets of inference. It follows that all relevant constraints imposed by the model are linear.

Examples of graphs in this class include the IV model in Figure 2 (a), as well as the models shown in Figures 3 (a) and (b). In particular, the model shown in Figure 3 (a) represents a scenario where we have separate instruments, A and B, for the true measurement X and the noisy proxy Y. To derive a set of linear constraints imposed by such a graph, we can generalize the procedure described for the IV model as follows. Let K be the number of variables in $\mathbf{X} = Ch(\Lambda)$. Then, for each variable $X \in \mathbf{X} = (X_i)_{i=1}^K$, let $\tilde{X}$ be the potential outcomes of X under each joint setting of its endogenous parents and let $\tilde{\mathbf{X}} = (\tilde{X}_i)_{i=1}^K$ be the set of all such potential outcomes. As before, let $\psi_{\tilde{\mathbf{X}}}$ denote the joint distribution of these potential outcomes. Recalling again that $\mathbf{A} \perp \tilde{\mathbf{X}}$, we have the equality $P(\mathbf{V} = \mathbf{v}) = P(\tilde{X}_1(\mathbf{v}_{Pa(X_1)}) = x_1, ..., \tilde{X}_K(\mathbf{v}_{Pa(X_K)}) = x_K)P(\mathbf{A} = \mathbf{a})$ where $\mathbf{V} = (\mathbf{X}, \mathbf{A})$ and $\mathbf{v}_{Pa(X_1)}$ are the values of X_1's parents in $\mathbf{v}$. As before, the right hand side is obtained through the (linear) marginalization of all other potential outcomes out of ψ, yielding linear constraints. Further, $P(\mathbf{A})$ is known, such that there are no other relevant constraints.

This class of graphs already contains several useful models, but it can still be expanded. We now present sufficient conditions for a graph $\mathcal{G}$ outside the class of instrumental graphs describe above to represent a model over endogenous variables equivalent to that of a graph $\mathcal{G}'$ inside the class. Any such $\mathcal{G}$ can be represented by the very same linear constraints as $\mathcal{G}'$, thereby extending the class of graphical models corresponding to linear constraints on $P(\mathbf{X} \mid \mathbf{A})$. This result is a special case of Proposition 5 in [5] and alternative proof is presented in the supplement.

Proposition 2. *Suppose a vertex A in a Bayesian network $\mathcal{G}$ has no observed or unobserved parents and has a single child B. Then the model for the variables in $\mathcal{G}$ is unchanged if an unobserved common parent of A and B is added to the graph, or if the unobserved common parent is added and the edge from A to B is removed.*

This proposition has two important consequences: first, instruments in $\mathbf{A}$ may be confounded with their children and, second, if an instrument is confounded with its child, it need not have a directed edge to that child. This result broadens the set of graphical models for which the constraints on the observed data distribution can be expressed linearly in ψ to include the graphs such as those shown in Figures 3 (c) and (d).

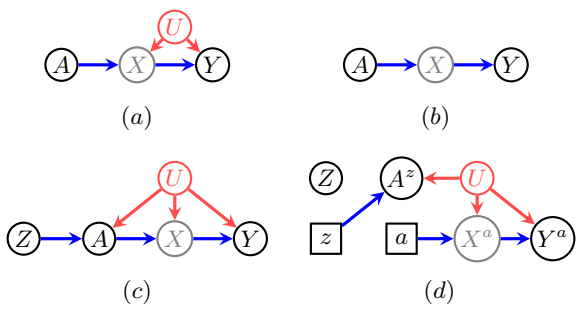

Figure 2: (a) - (c) show example Bayesian networks. Black variables represent observed variables, grey variables represent unobserved variables we wish to make inferences on, and red variables represent unobserved variables with unknown cardinality. (d) shows a single world intervention graph (SWIG) corresponding to graph (c) under the intervention $Z = z$ and $A = a$.

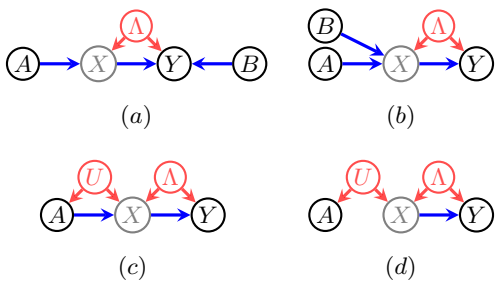

Figure 3: (a) and (b) are classical instrumental graphs as described in Proposition 1, (c) and (d) are graphs whose constraints on the observed data distribution can also be represented linearly by Proposition 2.

Once we have expressed the modeling constraints as linear constraints relating the parameters ψ to the observed data conditional distributions $P(\mathbf{x} \mid \mathbf{a})$, we can now proceed exactly as in Section 2, optimizing with respect to ψ rather than ϕ. Since ϕ is linearly related to ψ, the observed data constraint and all measurement error constraints from the previous section are still in linear in ψ and can be composed with the graphical constraints in this section. Alternatively, the measurement error constraints can be expressed for each potential outcome, representing a belief that these constraints hold in the observed data as well as under various interventions.

Application: Bounding $E[X]$ in the IV model

To demonstrate the use of graphical constraints, we will extend our example from Section 2 to include an additional binary instrument $A \in \{0, 1\}$ and we will assume the graphical model in Figure 2 (a). For a complete description of the resulting LP, see Appendix 1. Assume also that the observed conditional distributions $P(Y \mid A = a)$ for $a = 0, 1$ are the two marginal distributions shown in Figure 1 and that $P(A = 0) = \frac{1}{2}$. Then, using *only* the constraints encoded in the graph, we get the numerical bounds $E[X] \in [0.31, 4.69]$. These bounds can be made substantially tighter by including additional measurement error constraints, but it is worth noting that we can get non-trivial bounds by relying only on graphical constraints. □

3.1 Computing bounds for non-linear models

Unfortunately, many relevant models do not fall into the model class described above. In this section, we describe how non-sharp outer bounds can be derived for such cases. The only complete procedure for identifying all constraints implied by Bayesian networks on the distribution of a subset of their vertices is an application of quantifier elimination [9], which is infeasibly slow for many problems. When constraints are known to exist, for example by Evans' e-separation criterion [6], their exact form may not be known and may not be linear. When constraints are known, but are not linear, it may be possible to derive sharp bounds analytically. For example, the following proposition gives sharp bounds for a three variable Markov chain (Figure 2 (b)) over binary variables.

Proposition 3. *Let X and Y be binary variables such that $X \not\perp\!\!\!\perp Y$, let A be a discrete variable such that $A \perp\!\!\!\perp Y | X$ and $P(A = a) > 0$ for all a, and let $p_y = P(Y = y)$ and $p_{y|a} = P(Y = y \mid A = a)$. Then we have the following sharp bounds on $P(X = 1)$:*

$$P(X = 1) \in \bigcup_{y \in \{0,1\}} \left[\frac{p_y - \min_a p_{y|a}}{1 - \min_a p_{y|a}}, \frac{p_y}{\max_a p_{y|a}} \right] \quad (2)$$

However, such analytical bounds are not typically available. In these cases, *non-sharp* bounds can be derived for any graph by first repeatedly appealing to Proposition 2 and then adding adding a latent confounder that meets the criteria of Proposition 1. Specifically, any latent variable Bayesian network $\mathcal{G}$ can be converted to a new graph $\mathcal{G}'$ that meets the conditions of Proposition 1 through the following steps:

1. For any latent confounder U with exactly two children A and B such that $Pa(A) = \{U\}$ and $Ch(A) = \{B\}$ or $Ch(A) = \emptyset$, add an edge from A to B if it does not exist and remove U from the graph.

2. Add a latent confounder Λ, and an edge from Λ to each variable V in the graph for which $Pa(V) \neq \emptyset$.

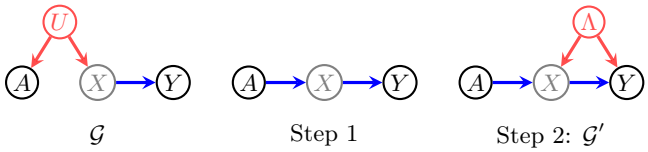

Figure 4: A non-linear graphical model $\mathcal{G}$ and an application of the linear relaxation procedure, resulting in the linear relaxed model $\mathcal{G}'$.

An example application of this procedure is shown in Figure 4. Because modifying the graph according to Proposition 2 does not change the constraints on $P(\mathbf{X}|\mathbf{A})$, Step 1 of this procedure does not change the constraint set. Further, adding an additional latent confound can only *remove* independencies from the graph, thus Step 2 represents relaxations of the constraints on ψ. As a result, applying the LP approach to the resulting model will result in *outer bounds* on the true partial identification set whose tightness will depend on how many edges were added in steps 2 and 3. In the following section, we extend our earlier discussion of potential outcomes to consider partial identification bounds for causal parameters and constraints relating multiple potential outcomes.

4 Causal parameters and constraints

In the previous section, we used potential outcomes to reason about the distribution of a mismeasured variable $P(X)$. Suppose instead that we are directly interested in the potential outcome $X(a)$ under some intervention $A = a$, but we are only able to observe X with measurement error. In this setting, our goal is to obtain bounds for parameters of the potential outcome distribution $P(X(a))$ for various settings a of A. This scenario is common in fields like economics and epidemiology, in which the treatment is exactly known, but the outcome is measured through inexact tools such as surveys. In this section, we will show how the constraints presented in the previous sections can be applied to target parameters involving potential outcomes and will introduce additional constraints that apply to causal inference settings. We demonstrate this approach in two important settings: a clinical trial with measurement error on the outcome, and an IV model with measurement error on the outcome.

We begin by showing how to map the distribution ψ to the distribution over arbitrary potential outcomes. Suppose that we assume a latent variable Bayesian network $\mathcal{G}$ that meets the conditions of Proposition 1 and that the joint distribution of instruments and potential outcomes ψ is defined as above. For an arbitrary potential outcome $X(a)$, we can linearly map ψ to $P(X(a))$ by setting the value of a and marginalizing out all mediators between X and A in $\mathcal{G}$, as in a structural equation model [16]. For example, assume the IV model in Figure 2 (a). In this case, the distribution $P(X(a))$ is just a marginal of ψ and the distribution $P(Y(a))$ can be derived as

$$P(Y(a) = y) = \sum_x P(Y(x) = y, X(a) = x) \tag{3}$$

In general, for endogenous variable X and intervention $A = a$, this expression can be constructed as follows. Let $\tilde{\mathbf{V}}$ represent all potential outcomes for variables other than X. The general expression is then

$$P(X(a) = x) = \sum_{\mathbf{v}} P(X(g(\mathbf{v}, A = a)) = x, \mathbf{V} = \mathbf{v}),$$

where $g(\mathbf{v}, A = a)$ is computed recursively as

$$g(\mathbf{v}, A = a) = \begin{cases} \mathbf{v}_{Pa(X)\setminus\{A\}}, A = a & \text{if } X \in Ch(A) \\ \{g(\mathbf{v}, C(a) = \mathbf{v}_{c(a)}) \text{ for } C \in Ch(A)\} & \text{otherwise.} \end{cases}$$

This marginalization is linear in ψ, and thus any causal parameter that can be written as a linear function of $P(X(a))$ can also be written as a linear function of ψ. This includes the average treatment effect (ATE) which is defined as $E[X(a) - X(a')]$ as well as the probability of non-zero treatment effect, which can be written as $P(X(a) \neq X(a'))$. As in the previous sections, we can express observed data, measurement error, and graphical constraints covered by Proposition 1 as linear constraints on ψ, allowing us to compute bounds on target parameters involving $P(X(a))$ under these constraints.

Applying Proposition 2 to this setting, however, requires a bit of care. Suppose we have a graph $\mathcal{G}$ in the instrumental class described by Proposition 1. If we are interested in the a potential outcome after intervening on one of the instruments, then adding a bidirected edge between that instrument and its child does, in fact, change the constraints on distribution of the potential outcome of interest. If, however, we are instead interested in the potential outcome of one child of Λ after intervention on another (i.e., we are not intervening on an instrument), then, in $\mathcal{G}$ as well as the post-intervention graph corresponding to this intervention, all instruments still satisfy the conditions of Proposition 2. Thus, any edge between an instrument and its child may be replaced or augmented with a bidirected edge.

The interesting consequence of this observation is that, so long as we are interested in a potential outcome after intervention on a child of Λ, Proposition 2 applies in

the causal setting, just as it does in the observational setting described above. Thus, for example, Figures 3 (c) and (d) lead to exactly the same bounds on $P(Y(x) = y)$ as does the classic IV model of Figure 2 (a). The bounds developed in [8] for models with traditional instruments can likewise be seen to apply to a larger set of graphs due to this result. Similarly, the measurement scenario depicted in Figure 2 (c), which is explored in detail below, would yield the same bounds on $P(X(a) = x)$ were the edge between Z and A replaced or augmented with a bidirected edge.

One intuition for partial identification results in the IV model is that variation in the treatment X *caused* by the instrument A is unconfounded with the outcome Y, which allows us to the effect of that part of the variation of X on Y. This proposition shows that the same reasoning applies when A and X are also, or only, confounded. Variation in X that participates in *covariation* with A is not confounded with the outcome Y, again allowing us to use that portion of the variation to partially identify the effect of X on Y.

Causal Assumptions

Finally, we may want to make **causal assumptions**, which relate potential outcomes under different interventions. For example, assuming we are interested in $X(a)$ and X is proxied by Y, below are two commonly made monotonicity assumptions which may be encoded as linear constraints.

(A5) Positive Effect of Treatment on Truth: $P(X(a) = x, X(a') = x') = 0 \; \forall a' > a, x' < x$

(A6) Positive Effect of Truth on Proxy: $P(Y(x) = y, Y(x') = y') = 0 \; \forall x' > x, y' < y$

(A5) is appropriate if there is are strong reasons to believe the outcome under one treatment with be strictly higher than under another. (A6) is employed whenever it is assumed that, even under measurement error, intervening to increase X will lead to an increase in Y. Additional causal constraints, such as limits on the effect size or the proportion affected, may be similarly imposed. As in the case of the measurement error assumptions, these equality constraints can be relaxed by specifying that the sums are bounded from above, rather than identically equal to zero.

Application: Bounding the ATE in a randomized trial

To demonstrate computation of bounds for a causal parameter, we will now extend our previous examples to compute bounds on the ATE in a randomized trial

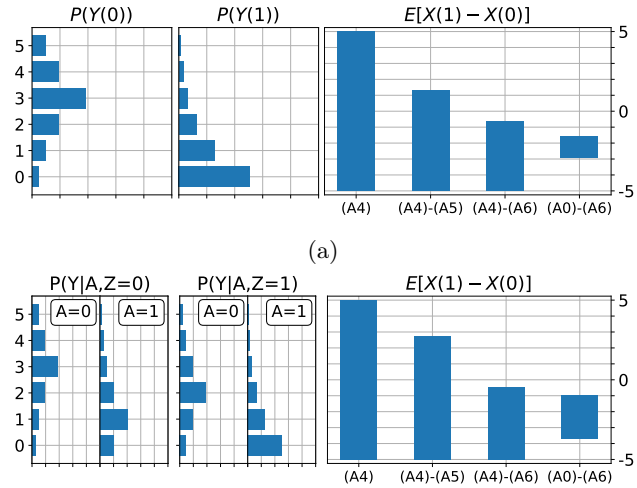

Figure 5: Bounds on the ATE $E[X(1) - X(0)]$ under different measurement error assumptions. (a) observed data distributions and bounds under in a randomized trial setting and (b) shows observed data distributions and bounds in an instrumental variable setting. Note that in the binary instrumental variable setting, Z can be thought of as the assigned treatment and A can be thought of as the observed treatment under partial compliance. $A = Z$ then corresponds to observed compliance and, in these cases ($A = Z = 0$ and $A = Z = 1$), the proxy distributions match the distributions in the randomized trial.

where there is measurement error on the outcome variable. Assume that we are interested in the ATE of a binary treatment $A \in \{0, 1\}$ on a variable X using data from a single noisy proxy Y, with X and Y defined as before. We will assume the graphical model shown in Figure 2 (a) where U represents an unobserved confounder. This model trivially satisfies the conditions of Proposition 1 and thus all graphical constraints can be expressed linearly. Figure 5 (a) shows the resulting bounds on the ATE as additional constraints are added. With only the exclusion restriction (A4), the bounds computed are the trivial bounds, though this will not be the case for all causal parameters of interest. With only the causal assumptions (A4)-(A6), however, we are able to meaningfully bound the ATE away from zero, though this bound becomes much tighter when the measurement error constraints are added. □

Application: Bounding the ATE in an instrumental variable model

Assume now that we are interested in the ATE of A on X, but that that A is no longer randomized. Instead

assume that we observe an additional binary variable $Z \in \{0, 1\}$ and that all variables follow the graphical model shown in Figure 2 (c) where again U represents an unobserved confounder. This model, which is a variation of the common instrumental variable (IV) model, similarly satisfies the conditions of Proposition 1 and we can again express all graphical constraints linearly. We assume that $P(A = 0 \mid Z = 0) = P(A = 1 \mid Z = 1) = 0.8$ and $P(Y \mid A, Z)$ is distributed as shown in Figure 5 (b). These distributions were chosen to roughly simulate a trial with partial compliance where Z represents treatment assignment and A represents actual treatment. The resulting bounds on the ATE under different assumptions are also shown in Figure 5 (b). As expected, these bounds are wider than those produced in the randomize trial case, but the ATE can nevertheless be meaningfully bounded away from zero with only the causal assumptions (A4)-(A6). □

We re-emphasize that no known symbolic bounds exist for either of these settings and that the various numerical bounds in each setting were computed by simply making small changes to the implied LP. For a full description of the LPs in both of these cases, see Appendix 4.

5 Related work

The ubiquity of measurement error in scientific study means that there is a vast literature on identification spread across a number of different methodological sub-disciplines. Much of this work concerns point identification in parametric models and we refer the interested reader to [3] and [10] for full treatments of these topics. In this section, we review several results on non-parametric identification in measurement error and related settings.

Several works, particularly in econometrics, have presented partial identifiability results under various measurement error models. [12] consider the setting presented in Section 2, deriving sharp bounds on the distribution of the ground truth under a particular error model where data is "contaminated" by data from another, unknown, distribution. [14] consider the same setting, presenting a procedure for verifying whether a particular distribution is in the identified set under a wide range of assumptions about the error distribution. [11] consider partial identifiability of a class of finite mixture models which includes as a special case the Markov chain model considered in Section 3, similarly proposing a method for verifying if a distribution is in the identified set. Our work differs from [14] and [11] in that our methods do not require guess-and-check to calculate the complete identified set.

The optimization-based approach we use to derive sharp bounds is inspired by the approach used by [1] to derive sharp bounds on causal effects in trials with partial compliance. This approach was similarly applied by [13] to partially identify the ATE under measurement error on the treatment variable. This work is also related to efforts to enumerate constraints on margins of latent variable Bayesian Networks implied by the model [18, 6, 4]. In such works, unobserved variables are not of primary interest and do not have known cardinality, so no attempt is made to bound functionals of their distribution. However, as indicated by our use of results from [2], constraints on the observed data law can be used to derive restrictions on unobserved variables of known cardinality.

6 Conclusions

In this work, we presented an approach for computing bounds on distributional and causal parameters involving a discrete variable which is subject to measurement error. At the heart of this approach is the encoding of the target parameter and modeling constraints as linear functions of the joint distribution of all variables in the model. The target parameter can then be maximized and minimized over the constraint set to produce sharp bounds for any observed data distribution. We provided a characterization of a class of graphical models that can be linearly expressed, and a procedure for finding a linear relaxation of models outside this class. An interesting consequence of this characterization is that randomized instruments that are confounded with treatments – rather than or in addition to causing them – can serve the same purpose as traditional instruments. This insight allows for the application of existing partial identification results to new models, and opens the door to analyses that make use of a wider variety of instruments in the future. We applied our approach to produce bounds under measurement error in settings with one or more proxies, including multiple important settings for which no known bounds currently exist.

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